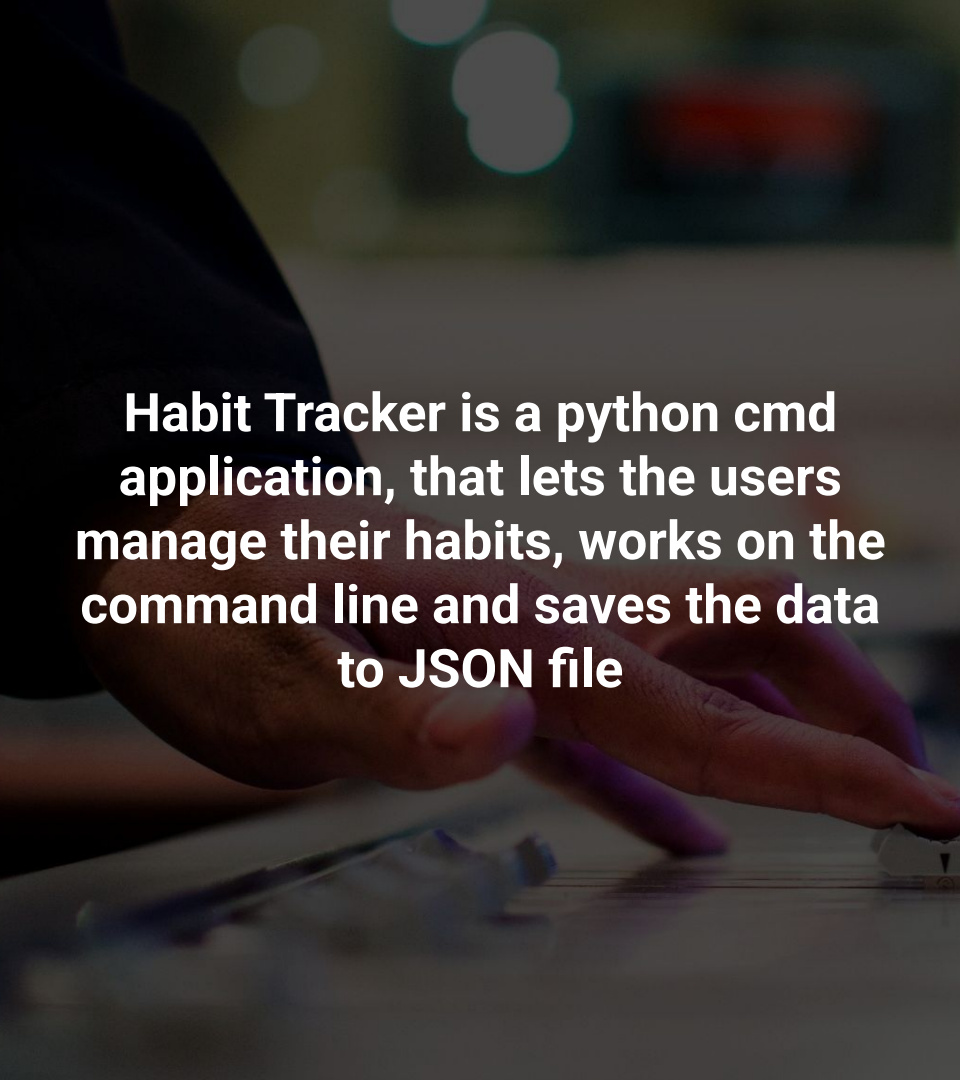


Habit Tracker cli App

Phase 2 Presentation

Object Oriented and Functional Programming with Python DLBDSOOFPP01

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Habit Tracker is a python cmd application, that lets the users manage their habits, works on the command line and saves the data to JSON file

1. Create daily and weekly habits with name and descriptions
2. Check off habits
3. View all habits
4. Delete a habit
5. See streaks for all habits
6. Comes with preloaded data for 4 weeks

What I used and why?

Python 3.10+ : Programming lang, Fast, and the syntax is easy to code with.

JSON : No database, easy to read and write

input() with loop : NO additional library installed, (used the native input mechanizm)

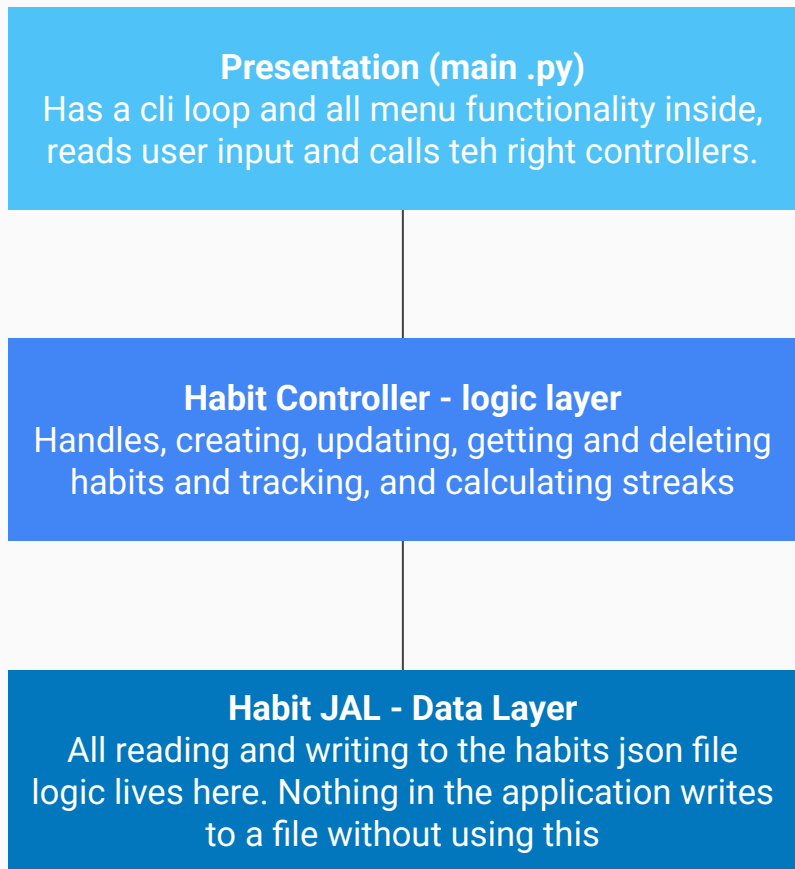
Filter, map, lambda : Functional programming modules, natively available, did not add any libraries.

Pytest : Very famous test framework for python, easy to use

Python-dateutil : helps work with time with different formats.

Architecture

3 layers, each with specific defined Jobs



Data Model

How the habit.json looks like

name : unique display name of the habit
description : short description of the habit

period : Daily or Weekly

created_at : timestamp of when the habit was created

completions : list of iso timestamps for each checked off habits

Core Features

What the cli python app do?

Create a Habit

User enters name, description and period (will check for duplicate name before saving)

Mark Completed

User picks a habit from the list, then append a new completion date

Delete Habit

User picks from a list, and can delete a habit, it deletes by name

Analytics

The app provides multiple analytics of the habits, like total streaks and streak by habit

Filter Habit

The user can view habits by filtering them using the period, (daily, weekly)

Storage

All data is stored to habit json file, if the file is corrupted, it will be restored

Analytics Module

All analytics functions use filter(), map(), and lambda so they are pure and do not modify any data. The analytics menu has 5 options

List all Habits

Lists all habits, with name, description and period concatenated

Filter Habit

The user can view habits by filtering them using the period, (daily, weekly)

Habit Streak

Show the streak for a specific habit the user picks

Every Habit Streak

Show the current streak for every habit

Longest Streak

Show which habit has the longest streak (daily and weekly separately)

Testing

The test suite uses pytest. Tests run against a separate JSON fixture (tests/habits_test.json) so they never touch the real habits.json.

Run with: `pytest tests/test.py -v`

- [weekly_create_habbit](#)

Creates a weekly habit and checks name, description, period are saved correctly

- [daily_create_habbit](#)

Same for a daily habit

- [weekly_mark_completion](#)

Marks the weekly habit done and checks the completions list grew

- [daily_mark_completion](#)

Same for a daily habit

- [get_all_habbit](#)

Checks exactly 2 habits exist after the two creates (in the test you create 2 habits)

- [daily_delete_habbit](#)

Deletes the daily habit and confirms it is gone

- [weekly_delete_habbit](#)

Deletes the weekly habit and confirms clean state

- [change_name](#)

Tests if change name for the username works

Installation and Usage

How to install and test the project

Step 1 - Install dependencies

Go into the Phase 2 folder and run: `pip install -r requirements.txt`

Step 2 - Start the app

Run: `python main.py`

Step 3 - Use the menu

Type a number (1-9) and press Enter. Type exit at any prompt to quit.

Step 4 - Run the tests

Run: `pytest tests/test.py -v`

An aerial photograph of the New York City skyline at dusk. The Empire State Building is prominently featured in the center, its top illuminated with red and green lights. The city is densely packed with skyscrapers, many of which have their lights on. The sky is a mix of dark blue and orange, indicating the time is either sunset or sunrise. The water of the harbor is visible in the distance.

Thank you

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